Experiment No: 14

Title: Configure Jenkins Nodes and Assign Jobs

Aim:

To configure Linux and Windows nodes in Jenkins and assign jobs to respective nodes for execution.

Objective:

- To understand distributed builds in Jenkins
- To configure Linux and Windows nodes
- To assign jobs to specific nodes
- To execute platform-specific builds

Prerequisites:

- Jenkins installed and running
- Java installed on master and nodes
- SSH access for Linux node
- Windows system with Jenkins agent setup

Tools Required:

- Jenkins
- Linux Machine
- Windows Machine
- Java (JDK)
- SSH Plugin

Theory:

Jenkins supports distributed builds using a **master-agent architecture**. The master node manages jobs, while agent nodes execute them.

A **Linux node** is connected using SSH, allowing Jenkins to run commands remotely. A **Windows node** is connected using an agent (JNLP or service).

This setup allows execution of platform-specific jobs, such as Java builds on Linux and MSBuild jobs on Windows.

Procedure:

Level 1a: Add Linux Node

1. Install SSH plugin in Jenkins
2. Go to **Manage Jenkins → Nodes → New Node**
3. Enter node name (Linux-node)
4. Select **Permanent Agent**
5. Configure:
    - Remote root directory
    - Labels: linux
    - Launch method: **Launch via SSH**
    - Provide IP, username, password
6. Save and connect

Level 1b: Add Windows Node

1. Go to **Manage Jenkins → Nodes → New Node**
2. Enter node name (Windows-node)
3. Select **Permanent Agent**
4. Configure:
    - Remote directory
    - Labels: windows
    - Launch method: **Launch agent via Java Web Start (JNLP)**
5. Run agent on Windows system
6. Connect node

Level 2a: Assign Java Job to Linux Node

1. Create Freestyle Job
2. Add label: linux

3. Add build step: Execute Shell
4. Add Java command or Maven build
5. Build job

Level 2b: Assign MSBuild Job to Windows Node

1. Create Freestyle Job
2. Add label: windows
3. Add build step: Execute Windows Batch Command
4. Add MSBuild command
5. Run job

Result:

Jobs were successfully assigned and executed on Linux and Windows nodes respectively.

Conclusion:

Jenkins distributed build system was successfully configured using Linux and Windows nodes. Jobs were executed on specific nodes based on their labels.

Level 1a → Add Linux Node:

Step 1 Actions

Go to Jenkins:

 Manage Jenkins → Nodes → New Node

Fill:

- Name: `linux-node`
- Select: **Permanent Agent**

Step 2 Configuration

Fill like this:

- Remote root directory:

`/home/your-username/jenkins-node`

- Labels:

```
linux
```

Step 3 Launch Method

 Choose:

Launch agent by connecting it to controller (JNLP)

Click Manage jemkins → linux_node

Step 1: Open Terminal

Step 2: Run FIRST command

```
curl -sO http://localhost:8080/jnlpJars/agent.jar
```

This downloads `agent.jar`

Step 3: Run SECOND command

Copy exactly from Jenkins (VERY IMPORTANT)

```
java -jar agent.jar -url http://localhost:8080 -secret YOUR_SECRET -name
"linux_node" -webSocket -workDir "/home/nasima/jenkins-node"
```

Expected Result

In terminal you should see:

```
Connected
```

Level 2a: Assign Job to Linux Node

Step 1: Open your Maven Job

 Go to:

Dashboard → your job (maven1 or Maven-Job)

Step 2: Configure Job

 Click:

Configure

□ Step 3: Assign to Linux Node

Find:
□ **General section**

□ Tick:

```
Restrict where this project can be run
```

In the box write:

```
linux
```

□ This must match the **label you gave earlier**

□ Step 4: Check Build Step

Make sure:

- Build step:
  □ **Invoke top-level Maven targets**
- Goals:

```
clean install -DskipTests
```

□ Step 5: Save

□ Click **Save**

□ Step 6: Build Now

□ Click:
Build Now

□ Step 7: Verify it ran on Linux Node

□ Click build → **Console Output**

Look for line like:

```
Running on linux_node
```

Expected Result

- Build runs successfully
- Status: **BUILD SUCCESS**
- Executed on: **linux_node**

Level 2b: Windows Node + MS Build

□ Step 1: Add Windows Node

Go to:
□ **Manage Jenkins → Nodes → New Node**

- Name: `windows_node`
- Type: **Permanent Agent**

□ Step 2: Configure

- Remote directory:

```
C:\jenkins-node
```

- Label:

```
windows
```

- Launch method:
  □ **Launch agent via JNLP**

□ Step 3: Run Agent on Windows

On Windows CMD:

```
curl -O http://<your-ip>:8080/jnlpJars/agent.jar
java -jar agent.jar -url http://<your-ip>:8080 -secret XXXXX -name windows_node
```

□ Step 4: Create Job

- New Item → `Windows-Job`
- Freestyle Project

□ Step 5: Assign Node

□ Tick:
Restrict where this project can be run

Write:

```
windows
```

□ Step 6: Add Build Step

□ Select:
Execute Windows batch command

Example:

```
echo Hello from Windows Node
dir
```

□ Step 7: Build

□ Click **Build Now**